

EC2010

Macroeconomics: Money and Monetary Policy

Summer 2023 Examination · Paper 1
Duration 1.5 hours · Section A: 4 of 6 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2010 — Macroeconomics: Money and Monetary Policy
Session	Summer 2023
Structure	Section A: six short questions, answer any four (50% of the paper between them). Section B: two long questions, answer one (50%) — Question B2 offers a further choice between part (b) and part (c) after the compulsory part (a).
This scheme covers	All six Section A questions, both Section B questions, and both the (b) and (c) alternatives within B2.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to the other EC2010 papers. A5(a) (the reaction function) is identical to the Summer 2024/2025 papers, just reweighted (60 marks here vs. 100 there). A6 (energy shock), B1(a) (Central Bank money creation) and B2(a) (transmission mechanism) cover the same underlying models as the other papers from a different angle. A1, A2, A3, A4, A5(b) and B2(b)/(c) are distinct to this sitting.

Section A · Question 1

Business Risk Perception & Bank Securities Purchases · 2 × 50 marks

A1**100 marks**

Explain how each of the following would be expected to affect the aggregate money supply: (In your answer you should show on the balance sheet how each affects the assets and liabilities of the commercial bank.)

- (i) Perception of a less risky business environment.
- (ii) Banks decide to purchase more government securities from the domestic (non-bank) public to build up their liquid assets.

FULL MODEL ANSWER**(i) A less risky business environment — 50 marks**

The mechanism: when banks perceive the business environment as less risky — lower expected probability of default across their loan book — two things happen together. Banks are willing to lend to a wider pool of borrowers at a given interest rate (looser credit standards, less need for capital buffers against risk-weighted assets), and would-be borrowers, facing a more confident economic outlook themselves, are more willing to take on new debt to invest or consume.

Result: loan applications that would previously have been declined (or priced prohibitively) are now approved. Each additional loan creates a new deposit at the moment it is drawn down (see Q. B1(a)), exactly as in the standard money-creation mechanism.

Commercial bank balance sheet	Assets	Liabilities
Additional loans approved due to lower perceived risk	+X	—
New borrower deposits created	—	+X
Net effect	+X	+X

The aggregate money supply **rises**, driven by a genuine loosening of lending standards rather than any change in Central Bank policy — this is exactly the kind of shift a central bank has to watch for in its data-dependent monitoring of credit conditions (see Q. A2/A5(b)).

(ii) Banks buy government securities from the domestic non-bank public — 50 marks

This is a more subtle case, and a common source of confusion: money creation is not limited to lending. When a bank buys a government bond from a non-bank seller (e.g. a household, a pension fund, a non-financial firm), it pays for the bond by crediting the seller's deposit account — not by handing over pre-existing reserves to the seller directly. A brand-new deposit is created, exactly as with a loan.

Commercial bank balance sheet	Assets	Liabilities
Government securities purchased	+X	—
Seller's deposit account credited	—	+X
Net effect	+X	+X

The aggregate money supply **rises** by the value of the purchase — the bank has expanded its balance sheet by acquiring an asset from outside the banking system,

and financed it by creating new deposit money, not by running down existing liquid assets.

Contrast worth stating explicitly: if the bank had instead bought the same securities from another bank, the transaction would just swap reserves for securities between banks in the interbank market — no new deposit would be created, and the money supply would be unaffected. The identity of the seller (bank vs. non-bank) is what determines whether the transaction creates new money. This is precisely the point emphasised in the Bank of England's 'Money Creation in the Modern Economy' reading: banks create money whenever they acquire an asset from — or extend credit to — a non-bank counterparty, not only through loans narrowly defined.

Mark Allocation — Part (i) (50 marks)

Tier	Marks	What separates this tier
40-50	Correctly identifies that lower perceived risk loosens both credit supply and demand, gives a fully correct balance sheet entry, and states the direction (money supply increases) clearly.	
25-39	Correct direction with a reasonable balance sheet entry, but the supply/demand mechanism is only partially explained.	
10-24	States the direction with little supporting mechanics.	
0-9	Incorrect direction or no meaningful treatment.	

Mark Allocation — Part (ii) (50 marks)

Tier	Marks	What separates this tier
40-50	Correctly identifies that buying securities from a non-bank creates a new deposit (money), with a fully correct balance sheet entry, and explicitly contrasts this with a purchase from another bank (no money creation).	
25-39	Correct direction and balance sheet entry, but doesn't draw out the non-bank vs. bank counterparty distinction.	
10-24	States that the money supply is affected with little supporting mechanics or an incorrect balance sheet entry.	
0-9	Incorrect direction (e.g. treats this as money-neutral) or no meaningful treatment.	

Question A1 Total: 100 Marks

Section A · Question 2

The Three Key ECB Rates & the Floor System · 3×25 + 25 marks

A2

100 marks

Read the following extract of the ECB's Monetary Policy Decisions, Feb 3rd 2023: "The Governing Council decided to raise the three key ECB interest rates by 50 basis points. Accordingly, the interest rate on the main refinancing operations and the interest rates on the marginal lending facility and the deposit facility will be increased to 3.00%, 3.25% and 2.50% respectively, with effect from 8 February 2023."

- (i) Explain each of the three key ECB interest rates.
- (ii) Identify the characteristics of the "Floor system".

FULL MODEL ANSWER

(i) The three key ECB interest rates — 3 × 25 marks

Rate (Feb 2023)	Level	What it is
Marginal lending facility (MLF)	3.25%	The ceiling. A standing facility offering banks overnight credit against eligible collateral whenever they fall short of liquidity; no bank borrows overnight above this rate, since it can always access this facility instead.
Main refinancing operations (MRO)	3.00%	The reference/signalling rate on the Eurosystem's regular (weekly) liquidity-providing open market operations, conducted as fixed-rate tenders with full allotment; historically the 'headline' policy rate.
Deposit facility (DFR)	2.50%	The floor. A standing facility letting banks deposit surplus overnight liquidity with the Eurosystem at a pre-set rate; no bank lends to another bank overnight below this rate, since it can always earn this rate risk-free instead.

The corridor: MLF (3.25%) sits 25bp above MRO (3.00%), and MRO sits 50bp above DFR (2.50%) — the same widened, asymmetric 50bp/25bp corridor structure introduced from July 2022 (see the equivalent question on the Summer 2024 paper), designed to reduce the appeal of parking cheap TLTRO-funded liquidity back at the deposit facility.

(ii) Characteristics of the "Floor system" — 25 marks

- Reserves in the banking system are **abundant** (well above the aggregate minimum reserve requirement), typically the result of large-scale asset purchases (APP/PEPP) and/or fixed-rate full-allotment refinancing operations.
- Because reserves are abundant, banks have little need to borrow from each other overnight to square their positions — a bank with surplus reserves has nowhere better to place them than the deposit facility.
- The overnight market rate (€STR) consequently trades very close to the deposit facility rate, not the MRO rate — the deposit rate becomes the operative policy rate in practice, even though the MRO rate is nominally 'the' policy rate.
- The width and symmetry of the interest rate corridor matters much less for short-term rate control than under a corridor system with scarce reserves, since the market rate gravitates to the floor regardless.
- This is the opposite regime to the pre-2008 'corridor system', where scarce reserves forced continuous active interbank trading and kept the market rate anchored near the middle of the corridor (the MRO rate) instead.

Link to the wider course: this is exactly the transition analysed in depth in the equivalent Summer 2025 question on the Eurosystem operational framework — if you're comparing schemes, that answer gives the full 'why' behind each of the bullet points above.

Mark Allocation — Part (i) (3 × 25 marks)

Tier	Marks	What separates this tier
20-25 each	Correctly identifies the rate's role (ceiling / reference rate / floor) and its current level, with a clear explanation of the mechanism (e.g. 'no bank lends below the floor because...').	
10-19 each	Correct basic identification of the rate but the explanation of its function in the corridor is thin.	
0-9 each	Rates confused with one another, or minimal explanation.	

Mark Allocation — Part (ii) (25 marks)

Tier	Marks	What separates this tier
20-25	Correctly identifies abundant reserves as the cause, explains why banks no longer need the interbank market, and correctly states that the market rate settles at the deposit facility rate rather than the MRO rate.	
10-19	States that the market rate sits near the floor without correctly explaining why (i.e. the reserve-abundance mechanism).	
0-9	Minimal or incorrect description.	

Question A2 Total: 100 Marks

Section A · Question 3

What Counts as Money? · 2 × 50 marks

A3**100 marks**Are each of the following “money”? Explain each answer.

- (i) Deposit accounts.
- (ii) Bitcoin.

FULL MODEL ANSWER

Economists judge whether something is “money” against the classic **functions of money** — medium of exchange, unit of account, store of value, and standard of deferred payment — and against practical properties like acceptability, stability of value, divisibility, and liquidity.

(i) Deposit accounts — Yes — 50 marks **Deposit accounts are money**

- **Medium of exchange:** deposits are used directly to settle transactions (debit cards, transfers, standing orders) — overwhelmingly the dominant means of payment in a modern economy.
- **Unit of account:** deposit balances are denominated in, and prices are quoted in, the same currency unit as cash.
- **Store of value:** deposits hold their nominal value over time (subject to inflation eroding real value, the same as cash) and are protected up to statutory limits by deposit guarantee schemes.
- **Standard of deferred payment:** contracts, loans and wages are routinely denominated and settled in deposit balances.
- **Officially recognised:** deposits are the largest component of standard monetary aggregates (M1 includes overnight deposits; M3 includes broader deposit types), are convertible into central bank money (cash) on demand, and are created and destroyed by regulated, centrally-backed commercial banks (see Q. A1, B1(a)).

Deposit accounts satisfy every function of money about as well as physical cash does, and are in fact the larger component of the money supply in practice.

(ii) Bitcoin — Generally No (or, at best, partial/contested) — 50 marks**Bitcoin fails most functions of money on conventional criteria**

- **Medium of exchange:** very limited acceptability — few merchants accept it directly, and where it is used, transactions are often routed through conversion to a fiat currency at the point of sale, undermining its role as a direct medium of exchange.
- **Unit of account:** almost nothing is priced in Bitcoin — prices are set in euro/dollar and merely converted at the prevailing exchange rate, which is itself extremely volatile.
- **Store of value:** Bitcoin's price has historically been highly volatile (large swings within single months), which is precisely what disqualifies an asset as a reliable store of value — holding Bitcoin exposes the holder to significant risk of loss (or gain) in purchasing power over short horizons, unlike a deposit account.
- **Standard of deferred payment:** contracts and debts are essentially never denominated in Bitcoin for the same reason — nobody wants to owe or be owed a fixed number of Bitcoin, given how much its real value can move before the debt is settled.

- **Not included in official monetary aggregates:** Bitcoin is not counted in M1/M3 by the ECB or other major central banks, is not legal tender in the Euro Area, has no central issuer or lender-of-last-resort backing, and its supply is fixed by protocol rather than responsive to the economy's transactional needs the way bank deposits are.

A balanced answer should note the debate: some economists argue Bitcoin functions as a speculative asset or a niche medium of exchange in specific contexts (e.g. cross-border remittances, jurisdictions with unstable domestic currencies, or as legal tender in El Salvador since 2021) rather than money in the conventional macroeconomic sense used throughout this course. The strongest answers reach a clear verdict (No, or at most partial) while acknowledging *why* the debate exists, rather than either dismissing Bitcoin outright or treating the question as unresolved.

Mark Allocation — Part (i) (50 marks)

Tier	Marks	What separates this tier
40-50	Correct verdict (yes) with a systematic run through the functions of money, correctly applied to deposit accounts.	
25-39	Correct verdict with some functions covered but not a systematic treatment.	
10-24	Correct verdict with minimal justification.	
0-9	Incorrect verdict or no meaningful justification.	

Mark Allocation — Part (ii) (50 marks)

Tier	Marks	What separates this tier
40-50	Correct verdict (generally no / partial) with a systematic run through the functions of money applied to Bitcoin, explaining specifically why volatility and limited acceptability undermine each function, and acknowledging the genuine debate.	
25-39	Correct verdict with some functions covered (e.g. volatility as a problem) but not systematic.	
10-24	Correct verdict with minimal justification, or a verdict of 'yes' with reasonable but incomplete justification.	
0-9	No meaningful justification either way.	

Question A3 Total: 100 Marks

Section A · Question 4

Large-Scale Asset Purchases & the Theory of Liquidity Preference · 100 marks

A4

100 marks

Use the Theory of Liquidity Preference to examine the impact of Large Scale Asset Purchase Programmes on interest rates. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

The **Theory of Liquidity Preference** (Keynes) explains the equilibrium interest rate as the price that equates the supply of and demand for money (rather than the price that equates saving and investment, as in the loanable funds framework).

Money demand (MD) slopes downward in (interest rate, real money balances) space: the interest rate is the opportunity cost of holding wealth as money rather than interest-bearing assets, so as the interest rate rises, people want to hold less money and more bonds/other assets. **Money supply (MS)** is drawn as vertical, since it is set directly by the central bank rather than determined by the interest rate.

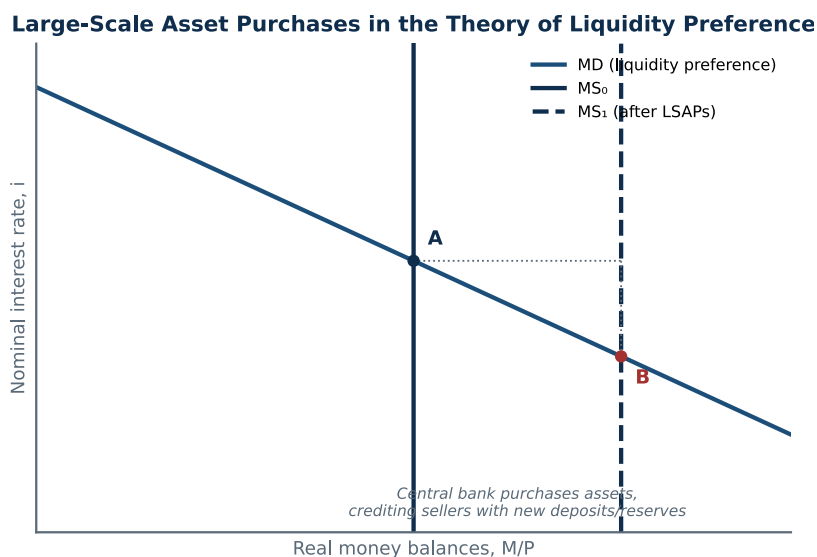


Fig. 1 — LSAPs shift the money supply right, lowering the equilibrium interest rate from A to B

How LSAPs enter the model

A Large Scale Asset Purchase Programme (e.g. the ECB's APP/PEPP) is the central bank buying bonds (usually government bonds) in the open market. When the seller is a non-bank (see Q. A1(ii)), payment is made by crediting the seller's bank with new reserves and the seller with a new deposit — this directly expands the money supply. Even when the seller is a bank, the central bank credits the bank's reserve account, again expanding the monetary base. Either way, **MS shifts right**, from MS_0 to MS_1 .

The resulting effect on interest rates

The liquidity effect: with MD unchanged and MS now further right, the new equilibrium (B) lies at a lower interest rate than the original equilibrium (A). Intuitively, the economy now has more money in circulation than before at the old interest rate, so asset holders bid down the return required to hold it (equivalently, bid up bond prices, which mechanically lowers bond yields) until the market clears again.

Beyond the basic liquidity-preference diagram — two reinforcing channels: LSAPs also work through channels the simple MD/MS diagram doesn't show directly: the portfolio balance effect (by removing a large quantity of long-term government bonds from the market, LSAPs push up the price of the remaining bonds and similar assets, lowering long-term yields specifically, not just short rates); and the signalling channel (a large asset purchase programme signals the central bank's commitment to an accommodative stance for an extended period, lowering expected future short rates and hence current long-term rates via the expectations hypothesis of the yield curve).

Exam tip: the question asks specifically for the Theory of Liquidity Preference, so the MD/MS diagram above should be the centrepiece of the answer — the portfolio balance and signalling channels are valuable additional depth but shouldn't replace the core liquidity-preference mechanism.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct MD/MS diagram with MS shifting right and the equilibrium rate falling, a clear explanation of why (money creation from the asset purchase, cleared by a lower required return on money), and additional discussion of the portfolio balance and/or signalling channels.	
50-74	Correct diagram and basic liquidity-effect explanation, but no discussion of the additional transmission channels beyond the simple MD/MS mechanism.	
25-49	Identifies that rates fall but the model/diagram is not correctly constructed (e.g. MS drawn sloping, or MD/MS roles confused).	
0-24	Minimal or largely incorrect treatment.	

Question A4 Total: 100 Marks

Section A · Question 5

The Reaction Function & Monetary Policy Stances · 60 + 40 marks

A5(a)**60 marks**

The following equation represents the reaction function of a Central Bank:

$$i = i^n + 1.5(\pi - \pi^T) + 0.5(Y - Y^T)/Y^T$$

where i^n = Natural (Neutral) Rate of Interest, π^T = Inflation Target, Y^T = Trend Output. Using this reaction function, explain how the Central Bank changes interest rates.

A5(b)**40 marks**

Distinguish between accommodative, neutral and contractionary monetary policy.

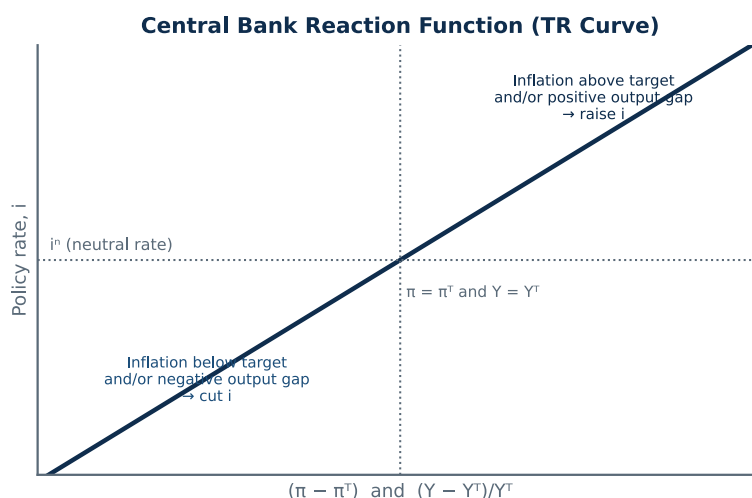
FULL MODEL ANSWER**(a) Interpreting the reaction function — 60 marks**

Fig. 1 — The TR curve implied by this reaction function

Inflation gap, $1.5(\pi - \pi^T)$: if inflation is above target, this term is positive and the bank raises the policy rate more than one-for-one (coefficient 1.5) — the Taylor principle, needed to actually raise the real rate and cool demand.

Output gap, $0.5(Y - Y^T)/Y^T$: if output is above trend, this term is positive and the bank raises rates by less per percentage point of gap (coefficient 0.5) than it does for an equivalent inflation gap.

Baseline: when $\pi = \pi^T$ and $Y = Y^T$, both terms vanish and $i = i^n$ — the bank simply holds the neutral rate.

(b) Accommodative, neutral and contractionary policy — 40 marks

These three stances describe where the actual policy rate sits relative to the neutral rate implied by the reaction function above (i.e. relative to i^n , adjusted for where inflation and output currently stand).

Stance	Condition (real rate vs. neutral)	Effect on the economy
Accommodative (expansionary)	Policy rate below the level the reaction function/neutral rate would imply	Stimulates demand: cheaper borrowing, lower saving returns, boosts output and, with a lag, inflation
Neutral	Policy rate equal to the level the reaction function/neutral rate implies	Neither stimulates nor restrains demand relative to its trend path; consistent with output at trend and inflation at target
Contractionary (restrictive)	Policy rate above the level the reaction function/neutral rate implies	Restrains demand: more expensive borrowing, higher saving returns, cools output and, with a lag, inflation

Practical difficulty: the neutral rate i^n itself is not directly observable — it must be estimated, and estimates are revised over time and disputed among economists. This means a central bank (and outside observers) can be genuinely uncertain in real time whether a given nominal rate is actually accommodative, neutral, or contractionary, especially once inflation is factored in to get the real rate (see the Fisher Equation, and the discussion of policy uncertainty elsewhere in this course).

Link to the reaction function: a central bank following the rule in part (a) is, by construction, accommodative whenever inflation is below target and/or output is below trend (both bracket terms negative, pulling i below i^n), and contractionary in the opposite case — the reaction function is really just a formal rule for choosing the appropriate stance given where the economy currently sits.

Mark Allocation — Part (a) (60 marks)

Tier	Marks	What separates this tier
45-60	Correctly explains both gap terms (relative magnitude of 1.5 vs 0.5), the Taylor principle, and the baseline case.	
30-44	Explains that the bank responds to each gap but misses the Taylor principle or the baseline case.	
15-29	General description without engaging with the coefficients.	
0-14	Minimal or incorrect interpretation.	

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
30-40	Correctly defines all three stances relative to the neutral rate, explains the effect of each on demand/inflation, and notes the practical difficulty of observing the neutral rate in real time.	
20-29	Correctly defines the three stances and their general effects, but no discussion of the neutral-rate measurement problem.	
10-19	Vague description of 'loose' vs 'tight' policy without a clear neutral-rate benchmark.	
0-9	Minimal or incorrect treatment.	

Question A5 Total: 100 Marks

Section A · Question 6

An Energy Shock: The AD/AS Model · 100 marks

A6

100 marks

Using an appropriate model, examine the impact of an energy shock on economic activity and prices. You should use a diagram to support your answer.

FULL MODEL ANSWER

The question doesn't specify a direction, so this answer works through the historically relevant case for this exam sitting — a rise in energy prices (an adverse energy shock, as experienced through 2021-2022) — and notes explicitly how the analysis reverses for a fall in energy prices.

Setting up the model

The appropriate tool is the **AD/AS model** in (output, inflation) space. The AD curve (downward sloping) is derived by combining the IS and TR relationships (see Q. A5). The AS curve (upward sloping in the short run) reflects firms' price-setting behaviour; energy is a direct input cost that shifts the whole curve.

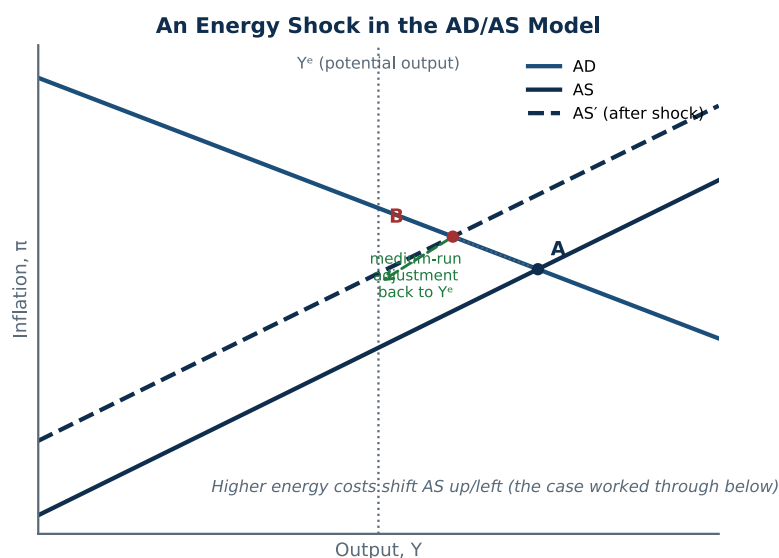


Fig. 1 — A rise in energy prices shifts AS left; the economy moves from A to B, then adjusts back towards Y^e over the medium run

Step 1 — The short-run impact of a rise in energy prices

A rise in energy prices directly increases firms' input costs. At any given level of output, firms need to charge a higher price to cover costs — this shifts the **AS curve up/left**. AD is unaffected in the first instance.

Result: the new short-run equilibrium (B) has higher inflation and lower output — stagflation. Output and inflation move in opposite directions, the signature of a supply-side (rather than demand-side) shock.

Step 2 — The medium-run adjustment

Output below potential eventually puts downward pressure on wages and non-energy costs, gradually shifting AS back towards its original position, provided the shock does not repeat and inflation expectations stay anchored. In the medium run output returns to Y^e and inflation returns towards target, but the price level is

permanently higher than it otherwise would have been.

The policy trade-off: if the central bank tightens in response to the inflation spike (raising i along TR), this shifts AD left too, bringing inflation down faster but pushing output further below potential — the defining dilemma facing a central bank after an adverse supply shock.

The mirror-image case: a fall in energy prices

A fall in energy prices works identically but with every sign reversed: AS shifts down/right, the short-run outcome is lower inflation and higher output (the opposite of stagflation — a genuinely favourable outcome, not a policy trade-off), and output above potential in the medium run puts upward pressure on wages, gradually shifting AS back as the temporary cost saving is absorbed.

Diagram guidance: a single AD/AS panel is sufficient. If your version of this question (or a related one on another paper) specifies the direction of the shock, use that direction — the mechanism is identical either way, just mirrored.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct AD/AS diagram, correctly identifies the short-run outcome (stagflation for a price rise, or the mirror-image for a price fall), and explains the medium-run adjustment back towards potential output, including the policy trade-off for an adverse shock.	
50-74	Correct diagram and short-run outcome, but the medium-run adjustment or policy trade-off is underdeveloped.	
25-49	Recognises the general direction of the effect but the model/diagram is not correctly constructed.	
0-24	Minimal or largely incorrect treatment.	

Question A6 Total: 100 Marks

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full, including both alternatives within B2

Section B questions are broader and typically integrate several ideas together. Question B2 offers a further choice: part (a) is compulsory, then you answer either part (b) or part (c). Both alternatives are covered in full below.

Section B · Question 1(a)

How the Central Bank Affects Money Creation, Two Ways · 2 × 25 marks

B1(a)**50 marks**

How does the Central Bank affect money creation? You should demonstrate your understanding by showing two different ways in which the Central Bank can affect the money supply. You should clearly show the relevant transactions on both the balance sheet of the commercial banks and the Central Bank.

FULL MODEL ANSWER

Two genuinely distinct channels are worked through below: one indirect (via the price of credit) and one direct (via balance sheet operations). A strong answer should keep this distinction explicit rather than presenting two variations on the same mechanism.

Channel 1: Setting policy interest rates (indirect) — 25 marks

Mechanism: the Central Bank sets its policy rate, which moves money-market rates and, via the standing facilities corridor (Q. A2), the rates commercial banks charge on loans and pay on deposits. A lower policy rate lowers the cost of borrowing, raising loan demand; since new lending creates new deposits, more lending means more money created. A higher rate works in reverse.

Commercial bank balance sheet (after a rate cut raises loan demand)	Assets	Liabilities
Additional loans extended	+X	—
New borrower deposits created	—	+X
Net effect	+X	+X

Central Bank balance sheet	Assets	Liabilities
No direct balance sheet change from the rate decision itself	—	—
(Balance sheet changes only as banks subsequently draw more/less on refinancing operations)	±	±

Key feature: the Central Bank's own balance sheet doesn't necessarily expand directly from a rate decision alone — the effect on money creation runs through commercial banks' lending decisions, which is why this channel is described as indirect.

Channel 2: Large-scale asset purchases (direct) — 25 marks

Mechanism: the Central Bank buys securities (e.g. government bonds) directly in the open market (see also Q. A4). If the seller is a non-bank, the transaction mechanically creates new reserves at the seller's commercial bank and a new deposit for the seller — both balance sheets expand immediately and directly, with no need for any change in bank lending behaviour.

Central Bank balance sheet	Assets	Liabilities
Government bonds purchased	+X	—
Reserves credited to seller's commercial bank	—	+X
Net effect	+X	+X

Commercial bank balance sheet	Assets	Liabilities
Reserves at the Central Bank	+X	—
Seller's deposit account credited	—	+X
Net effect	+X	+X

Key feature: both base money (reserves) and broad money (bank deposits) rise by the purchase amount immediately — this channel is described as direct because the Central Bank's own balance sheet action is what creates the new money, not a subsequent lending decision by commercial banks.

What distinguishes the two channels: Channel 1 changes the incentives banks and borrowers face and relies on them responding (money creation is a consequence of others' decisions); Channel 2 is a mechanical, immediate balance sheet expansion that creates money the moment the transaction settles, regardless of subsequent lending behaviour. A strong answer states this distinction explicitly rather than just describing the two mechanics side by side.

Mark Allocation (50 marks, 25 per channel)

Tier	Marks	What separates this tier
20-25 per channel	Correct mechanism, fully correct balance sheet entries on both the commercial bank and Central Bank sides (where applicable), and a clear statement of whether the channel is direct or indirect and why.	
10-19 per channel	Correct mechanism with broadly correct balance sheet entries, but missing one side (e.g. Central Bank balance sheet omitted) or the direct/indirect distinction not made.	
0-9 per channel	Minimal or incorrect balance sheet treatment.	

Section B · Question 1(b)

Does Money Supply Growth Explain Price Inflation? · 50 marks

B1(b)**50 marks**

Does the growth in the money supply help to explain price inflation? Explain your answer. You should use the economic theory and relevant readings that you have studied in EC2010 to support your answer.

FULL MODEL ANSWER**The starting framework: the Quantity Theory of Money**

The Equation of Exchange, $MV = PY$ (see the equivalent question on other EC2010 papers), becomes the **Quantity Theory of Money** once velocity (V) is assumed roughly stable and real output (Y) is determined independently by real factors. Under those assumptions, money supply growth should translate fairly directly into price inflation over the medium-to-long run — the classical basis for treating inflation as “always and everywhere a monetary phenomenon.”

Where the theory fits the data well

The post-pandemic surge (2021-2022): Euro Area M3 growth accelerated sharply from 2020, driven by pandemic-era fiscal support financed substantially through PEPP purchases, flowing into household and firm deposits (furlough schemes, forced saving during lockdowns). As economies reopened, velocity rebounded (pent-up demand, release of accumulated savings) rather than offsetting the money growth as it had in the previous decade — M and V moved in the same direction, and headline inflation duly rose to double digits by late 2022. This episode is broadly consistent with the QTM: substantial money growth, reinforced rather than offset by velocity, and (with a lag) substantial inflation.

Where the theory needs qualification

The 2015-2018 counter-example: the ECB's APP from 2015 expanded the monetary base substantially, yet headline and underlying inflation remained persistently below target for most of this period — on a naive reading, a QTM failure. The resolution is that velocity fell: much of the new reserve money did not flow into broad money and spending at the same pace (weak credit demand, banks holding reserves rather than lending), so the identity $MV=PY$ still held, but the M -to- P link was much weaker in the short-to-medium run than a mechanical reading of the QTM would suggest.

Supply shocks played an independent role in 2021-2022: the energy price shock (Q. A6) and pandemic-related supply chain disruption pushed the AS curve left at the same time as monetary-driven demand pushed AD along a QTM-consistent path — so part of the 2021-2022 inflation surge reflects a genuine supply shock, not monetary growth alone. Isolating “how much of inflation was due to money growth” versus “how much was due to the energy shock” is genuinely difficult and disputed.

The composition of money creation matters: money created via reserves sitting on bank balance sheets (as in 2015-2018) has a much weaker link to spending and inflation than money created via deposits directly reaching households and firms who then spend it (as in 2021-2022) — the QTM's assumption of stable velocity implicitly assumes away this compositional difference.

Verdict

Yes, but only as one part of the story, and not mechanically

Money supply growth does help explain price inflation — it is close to a necessary condition for a sustained rise in the price level, and the broad correlation between accelerating M3 growth and the subsequent inflation surge in 2021-2022 supports this. But money growth alone is not sufficient to explain inflation's timing or magnitude in any given episode: velocity instability (as in 2015-2018) and independent supply shocks (as in 2021-2022) both mean the short-run relationship between M and P is loose and lagged, even though the underlying QTM mechanism is real and, per the Schnabel Thünen Lecture referenced elsewhere in this course, not a vacuous concept for modern central banks.

What a strong answer needs: explicit use of $MV=PY$, engagement with *why* the money-inflation link was weak in one period and strong in another (not just an assertion that it varies), and integration of the money-creation mechanics from Q. B1(a) and Q. A4 rather than treating the QTM as a black box.

Mark Allocation (50 marks)

Tier	Marks	What separates this tier
40-50	Correctly sets out $MV=PY$, explains both the 2021-2022 fit and the 2015-2018 counter-example via velocity, discusses the independent role of supply shocks and the composition of money creation, and reaches a clear, well-supported verdict.	
25-39	Uses the QTM framework and discusses at least one supporting and one complicating episode, but the explanation of velocity's role is thin.	
10-24	General discussion of money and inflation without meaningfully engaging with the Quantity Theory or $MV=PY$.	
0-9	Minimal or off-topic answer.	

Question B1 Total: 100 Marks (50 + 50)

Section B · Question 2(a)

The Monetary Transmission Mechanism: Channels · 70 marks (compulsory within B2)

B2(a)**70 marks**

Central Banks such as the ECB are using standard monetary policy instruments (interest rates) in response to higher price inflation (See Fig. 1). Use the economic models in EC2010 to uncover the monetary transmission mechanism, i.e. the channels through which increasing interest rates will affect price inflation.

FULL MODEL ANSWER

A complete answer identifies the individual channels through which a policy rate rise transmits to the real economy and prices, before summarising the overall effect using the IS-TR/AD-AS models. Each channel is a distinct economic mechanism — a strong answer keeps them separate rather than lumping them into one generic “higher rates reduce spending” statement.

Channel 1: The direct interest rate channel

Mechanism: a higher policy rate raises the cost of borrowing (for mortgages, business loans, credit) and raises the return to saving. Both effects discourage current spending relative to future spending — households defer durable purchases and firms defer investment projects with borrowing-cost-sensitive returns. This is the core mechanism behind the downward-sloping IS curve (Q. A4, A5).

Channel 2: The bank lending / credit channel

Mechanism: higher rates raise banks' own funding costs and, combined with a weaker growth outlook, typically lead banks to tighten credit standards independently of the pure price effect in Channel 1 — some borrowers who could obtain credit before now cannot at any price, not just at a higher price. Since new lending creates new deposits (Q. A1, B1(a)), a tighter lending channel directly slows money creation, reinforcing the interest rate channel.

Channel 3: The exchange rate channel

Mechanism: a higher domestic policy rate, relative to rates abroad, attracts capital inflows seeking the higher return, which tends to appreciate the domestic currency. An appreciation makes imports cheaper (directly reducing imported inflation, e.g. energy priced in dollars) and exports less competitive (reducing net exports, another drag on aggregate demand via the IS curve's NX component).

Channel 4: The asset price / wealth channel

Mechanism: higher rates lower the present value of future cash flows, reducing the prices of bonds, equities and (with a lag) housing. Lower asset prices reduce household and firm wealth, which reduces consumption and investment (a wealth effect operating alongside, and reinforcing, the direct interest rate channel).

Channel 5: The expectations / signalling channel

Mechanism: a rate rise (especially alongside forward guidance or a credible commitment to the inflation target — see Q. A2(b) on other papers) signals the central bank's determination to bring inflation back to target. If this anchors household and firm inflation expectations, it can directly moderate wage demands and price-setting behaviour, dampening inflation even before the demand-side channels above have fully worked through — this is precisely the mechanism behind why credible, well-communicated policy can reduce the output cost of disinflation.

Putting the channels together: the reduced-form IS-TR / AD-AS summary

All five channels above ultimately show up as: a higher policy rate shifts the **TR curve up** in the IS-TR model, reducing output in the short run (Channels 1-4 collectively determine how much output falls for a given rate rise); lower output then reduces inflationary pressure, shown as the **AD curve shifting left** in the AD/AS model, with Channel 5 (expectations) potentially also shifting the AS curve favourably if credibility is strong.

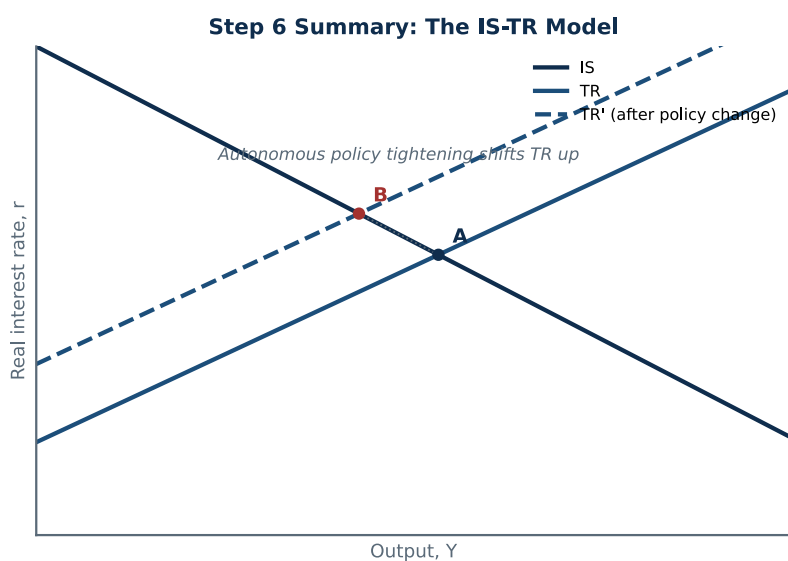


Fig. 1 — The combined effect of Channels 1-4 in the IS-TR model

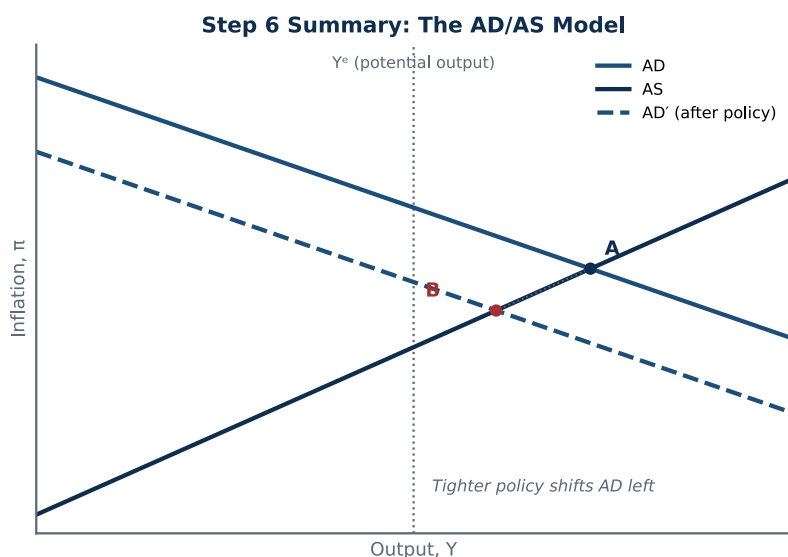


Fig. 2 — The resulting effect on inflation in the AD/AS model

Why 'channels' matters for the mark scheme: simply drawing the IS-TR/AD-AS diagrams (as in the equivalent question on the Summer 2024/2025 papers) captures the reduced-form outcome but not the individual mechanisms the question asks for here. Full marks require identifying and explaining each channel separately before summarising with the diagrams.

Mark Allocation (70 marks)

Tier	Marks	What separates this tier
55-70	Identifies and correctly explains at least four distinct channels (interest rate, bank lending/credit, exchange rate, asset price/wealth, expectations/signalling), and summarises the combined effect using the IS-TR/AD-AS models with correct diagrams.	
35-54	Identifies and explains two to three channels correctly, with a reasonable summary diagram.	
15-34	Only the basic interest rate channel is discussed, or channels are named without real explanation of the mechanism.	
0-14	Minimal or largely incorrect treatment.	

Section B · Question 2(b)

Challenges Setting Interest Rates: Rising vs. Falling Inflation · 30 marks (Alternative 1 within B2)

B2(b)

30 marks

With reference to Fig. 1, analyse the challenges that central banks such as the ECB face in setting interest rates during periods of rising and falling inflation.
Or

FULL MODEL ANSWER

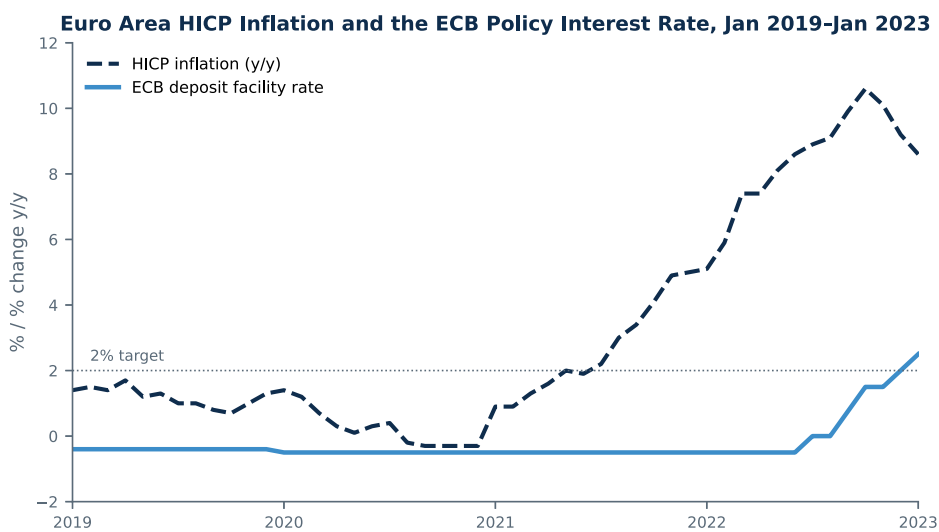


Fig. 1 — Illustrative reconstruction of the published path of Euro Area HICP inflation and the ECB policy rate, based on Eurostat/ECB data

Challenges during the rising phase (2021 through late 2022)

Distinguishing transitory from persistent inflation in real time: when inflation first accelerated in 2021, it was genuinely unclear (and disputed among central banks and forecasters) whether this was a temporary pandemic-reopening effect (supply chain bottlenecks, base effects) that would fade on its own, or the start of a more persistent, demand-driven episode requiring a policy response. Acting too early on what turns out to be transitory imposes needless output costs; acting too late on what turns out to be persistent lets inflation expectations de-anchor.

Transmission lags mean policy must be forward-looking, not reactive: because rate changes take 12-24 months to fully affect inflation (Q. B2(a)), the ECB has to set policy based on a forecast of where inflation will be, not where it currently is — and forecasting during a highly unusual, fast-moving episode (pandemic reopening, war-driven energy shock) was subject to unusually large errors.

A large part of the initial surge was a supply shock, not a demand shock: the AD/AS model itself (Q. A6) shows that raising rates in response to a supply shock is less effective, and more costly in output terms, than raising rates in response to a genuine demand shock — yet the interest rate instrument doesn't distinguish between the two, so the ECB faced a genuine dilemma about how hard to lean against an inflation surge it could only partially influence.

Risk of ‘falling behind the curve’: as the chart shows, the policy rate remained deeply negative in real terms (nominal rate far below inflation) for most of 2021-2022, raising credibility concerns and the risk that persistently negative real rates would let inflation expectations drift upward, requiring even larger rate rises later to re-anchor them.

Challenges during the falling phase (from late 2022 as inflation peaks and turns)

Knowing when disinflation is durable versus temporary: energy prices falling back can produce a rapid, mechanical decline in headline inflation that doesn't reflect underlying/core inflation dynamics (wages, services) — easing too early on a headline-driven fall risks a second wave of inflation if underlying pressures haven't actually eased.

The risk of overtightening given lags: because earlier rate rises are still working through the system with a lag, continuing to tighten aggressively as headline inflation falls risks compounding effects that weren't yet visible when each individual decision was made, potentially overshooting into an unnecessarily deep slowdown.

Asymmetric costs of error: cutting too soon (or stopping hikes too soon) risks a resurgence of inflation and a costlier second tightening cycle to re-establish credibility; continuing to hike too long risks an avoidable recession. Central banks must weigh these asymmetric risks explicitly, which is part of why the ECB adopted the explicitly data-dependent, meeting-by-meeting communication style seen in later ECB statements (see the equivalent A2(b)-style question on other EC2010 papers) rather than committing to a fixed path in either direction.

The common thread

Both phases share the same underlying difficulty: interest rates are a blunt, lagged instrument being deployed against a partially unobservable, fast-moving target (the future inflation path), with highly asymmetric and uncertain costs of error in either direction. Fig. 1 illustrates this concretely — the policy rate visibly lags the inflation curve by design (it responds to the inflation outlook, not just the latest print), which is exactly what generates the challenges analysed above.

What a strong answer needs: use Fig. 1 as evidence for specific claims (e.g. pointing to the size of the negative real rate gap in 2021-2022, or the continued rate rises after inflation had visibly turned) rather than only describing the chart's shape.

Mark Allocation (30 marks)

Tier	Marks	What separates this tier
22-30	Identifies distinct challenges for both the rising and falling phases (e.g. transitory-vs-persistent judgement, transmission lags, supply-shock origin, overtightening risk, asymmetric costs of error) and explicitly references Fig. 1 as evidence.	
14-21	Identifies challenges for at least one phase in reasonable depth, with some reference to Fig. 1.	
6-13	General discussion of central bank difficulty without distinguishing the two phases or engaging with Fig. 1.	
0-5	Minimal or off-topic answer.	

Question B2 (with option b) Total: 100 Marks (70 + 30)

Section B · Question 2(c)

Projecting the ECB Policy Rate for H2 2023 · 30 marks (Alternative 2 within B2)

B2(c)**30 marks**

Provide your projection for the ECB's policy interest rate for the second half of 2023. In your answer you should state clearly the assumptions that you are making in order to derive your answer.

FULL MODEL ANSWER

As with the equivalent H2 2024 question on the Summer 2024 paper, this is an open forecasting question: the mark scheme rewards clearly stated, economically sound assumptions applied consistently via the course's own models, not any single "correct" number. A strong answer as of early-to-mid 2023 might have looked like this.

Step 1 — State the starting point and the assumptions

- **Starting point:** deposit facility rate at 2.50% following the February 2023 decision (Q. A2), with headline HICP inflation still running well above target (roughly 8–9% year-on-year in early 2023, per Fig. 1) though visibly past its October 2022 peak.
- **Assumption 1 — underlying/core inflation remains sticky even as headline falls:** energy base effects will mechanically pull headline inflation down through 2023, but services inflation and wage growth (in a still-tight labour market) are assumed to remain elevated for longer, keeping underlying inflationary pressure above target.
- **Assumption 2 — the ECB prioritises credibility after being seen as slow to start hiking:** having been criticised for reacting late to the initial inflation surge, the Governing Council is assumed to want clear evidence that underlying inflation is durably on a path back to target before pausing, rather than stopping as soon as headline inflation improves.
- **Assumption 3 — the ECB continues to apply its own reaction function (Q. A5) and data-dependent approach:** further hikes remain likely as long as the inflation gap term in the reaction function stays materially positive, delivered in a sequence of smaller, more cautious steps than the 50-75bp moves seen in 2022, consistent with the gradualism/uncertainty logic discussed elsewhere in this course.

Step 2 — Apply the framework and state the projection

Projection: continued tightening through the first part of H2 2023, most likely in a series of 25bp increments (smaller than the 2022 pace, reflecting increased caution as the policy rate approaches restrictive territory), followed by a pause once the Governing Council judges the cumulative tightening — combined with the lagged effect of hikes already delivered — sufficiently restrictive to bring underlying inflation durably back towards target. A plausible end-2023 deposit facility rate in the region of 3.50–4.00%, with the pace of further hikes slowing markedly compared to 2022.

Key risks stated explicitly: a resurgence in energy prices or renewed supply chain disruption could require more tightening than this baseline; conversely, a sharper-than-expected growth slowdown or banking-sector stress (a live concern given banking-system turmoil in some other jurisdictions in early 2023) could justify an earlier pause than assumed here.

What actually happened (for study reference — postdates the exam, not required for full marks): the ECB delivered four further 25bp hikes — March, May, June, and July 2023, plus a final 25bp hike in September 2023 — taking the deposit facility rate from 2.50% to 4.00% by September 2023, then held rates unchanged for the rest of the year. This came in close to the upper end of the illustrative 3.50-4.00% baseline range above, and confirms the 'continued but slowing, then pause' pattern that applying the course's own reaction-function logic would have suggested.

Mark Allocation (30 marks)

Tier	Marks	What separates this tier
22-30	States clear, economically sound assumptions (underlying inflation trajectory, ECB credibility considerations, reaction-function behaviour), applies them consistently to derive a directionally sound projection (continued but slowing tightening, then a pause), and explicitly discusses risks in both directions.	
14-21	Provides a reasonable projection with some stated assumptions, but the link to the course's models is weak.	
6-13	States a number with minimal justification or assumptions.	
0-5	No meaningful projection or assumptions given.	

Marking note: as with the equivalent question on other papers, a well-reasoned answer that reaches a different numerical projection should score just as highly as one close to the actual outturn, provided the assumptions are stated clearly and applied consistently. Penalise unsupported assertions, not directional disagreement.

Question B2 (with option c) Total: 100 Marks (70 + 30)